MATH 342W / 642 / RM 742 Spring 2026 HW #4

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Problem 1

These are questions about Silver's book, chapters 7–11. You can skim chapter 10 as it is not so relevant for the class. For all parts in this question, answer using notation from class (i.e. $t, f, g, h^*, \delta, \epsilon, e, t, z_1, \ldots, z_t, \mathbb{D}, \mathcal{H}, \mathcal{A}, \mathcal{X}, \mathcal{Y}, X, y, n, p, x_{\cdot 1}, \ldots, x_{\cdot p}$, $x_{1\cdot}, \ldots, x_{n\cdot}$, etc.) as well as in-class concepts (e.g. simulation, validation, overfitting, etc) and also we now have $f_{pr}, h^*_{pr}, g_{pr}, p_{th}$, etc from probabilistic classification as well as different types of validation schemes).

Note: I will not ask questions in this assignment about Bayesian calculations and modeling (a large chunk of Chapters 8 and 10) as this is the subject of Math 341/343. We also won't cover chapters 12-13 and the conclusion on the homework.

(a) [easy] Why are flu fatalities hard to predict? Which type of error is most dominant in the models?

> Flu fatalities are hard to predict because the true process t depends on a lot of unknown and changing factors, like mutations and human behavior. Our model f can't capture all of that, and the data is noisy. The biggest error here is ignorance error δ.

(b) [easy] In what context does Silver define extrapolation and what term did he use? Why does his terminology conflict with our terminology?

> Silver uses "extrapolation" to mean extending trends beyond the data. In class, extrapolation is more specific: using a model outside the observed X. His definition mixes in overconfidence, which is why it conflicts with ours.

(c) [easy] Give a couple examples of extraordinary prediction failures (by vey famous people who were considered heavy-hitting experts of their time) that were due to reckless extrapolations.

Examples include experts saying housing prices would keep rising forever or financial models assuming markets were always stable.

(d) [easy] Using the notation from class, define "self-fulfilling prophecy" and "self-canceling prediction".

A self-fulfilling prophecy is when a prediction changes behavior so it becomes true: $t \approx g$.
A self-canceling prediction is when a prediction changes behavior so it does not happen: $t \neq g$.

(e) [easy] Is the SIR model of infectious disease under or overfit? Why?

The SIR model is underfit because it is too simple and cannot capture all the real-world complexity of disease spread.

(f) [easy] What did the famous mathematician Norbert Weiner mean by "the best model of a cat is a cat"?

Wiener meant that reality t is extremely complex, and no model f can fully match it. Models are always simplifications.

(g) [easy] Not in the book but about Norbert Weiner. From Wikipedia:

> Norbert Wiener is credited as being one of the first to theorize that all intelligent behavior was the result of feedback mechanisms, that could possibly be simulated by machines and was an important early step towards the development of modern artificial intelligence.

What do we mean by "feedback mechanisms" in the context of this class?

Feedback mechanisms mean outputs affect future inputs. In class terms, Y_t influences future X, so the system keeps updating itself.

(h) [easy] I'm not going to both asking about the bet that gave Bob Voulgaris his start. But what gives Voulgaris an edge (p239)? Frame it in terms of the concepts in this class.

Voulgaris has an edge because his model g is better than others. He uses better data and makes better predictions than the market.

(i) [easy] Why do you think a lot of science is not reproducible?

> A lot of science is not reproducible because researchers overfit their data D, test too many ideas, and don't properly validate their results, a topic he also talked about in chapter 2.

(j) [easy] Why do you think Fisher did not believe that smoking causes lung cancer?

> Fisher didn't believe smoking caused cancer because of his prior beliefs. He misinterpreted the data and didn't accept the causal relationship because it didn't fit his preconceived ideas.

(k) [easy] Is the world moving more in the direction of Fisher's Frequentism or Bayesianism?

> Silver emphasizes Bayesian thinking in Chapter 8, where he explains that good predictions come from updating beliefs as new data becomes available.

(l) [easy] How did Kasparov defeat Deep Blue? Can you put this into the context of over and underfitting?

> Kasparov was able to challenge Deep Blue by playing in ways the computer didn't expect. Deep Blue relied on a fixed model of chess, while Kasparov used intuition and strategies the computer didn't model well. The computer was powerful but overfit, unable to adapt.

(m) [easy] Why was Fischer able to make such bold and daring moves?

> Fischer was able to make bold moves because he had a very strong understanding of the game and trusted his judgment. His "model" was very accurate.

(n) [easy] What metric y is Google predicting when it returns search results to you? Why did they choose this metric?

> The metric y Google is predicting is whether users click on and are satisfied with the search results. They chose this metric because it is measurable and gives direct feedback about how useful the results are, allowing them to improve their predictions over time.

(o) [easy] What do we call Google's "theories" in this class? And what do we call "testing" of those theories?

> Google's "theories" are models in the hypothesis set $\mathcal{H}$. "Testing" those theories is validation, where we check how well the model performs on new data.

(p) [easy] p315 give some very practical advice for an aspiring data scientist. There are a lot of push-button tools that exist that automatically fit models. What is your edge from taking this class that you have over people who are well-versed in those tools?

> On page 315, Silver's advice is to think in terms of probabilities and update predictions based on new information. He suggests that good forecasting comes from understanding uncertainty, not just using tools. My edge from this class is that I know how to evaluate models and think critically about predictions instead of just relying on automated outputs.

(q) [easy] Create your own 2×2 luck-skill matrix (Fig. 10-10) with your own examples (not the ones used in the book).

	Low Luck	**High Luck**
High Skill	Playing piano	Stock Investing
Low Skill	Walking	Rolling a die

(r) [easy] [EC] Why do you think Billing's algorithms (and other algorithms like his) are not very good at no-limit hold em? I can think of a couple reasons why this would be.

> Billing's algorithms struggle with no-limit hold em because the game has hidden information and unpredictable human behavior. This makes it very hard to model accurately, so the algorithm cannot fully capture the true process.

(s) [easy] Do you agree with Silver's description of what makes people successful (pp326-327)? Explain.

> I mostly agree. Success comes from good decision-making and learning over time, but luck still plays a role and can affect outcomes.

(t) [easy] Silver brings up an interesting idea on p328. Should we remove humans from the predictive enterprise completely after a good model has been built? Explain

No, we should not remove humans. Models still have errors and limitations, and humans are needed to interpret results and adjust when things change. This goes back to the edge this class gives us, because the ability to interpret and think is still important.

(u) [easy] According to Fama, using the notation from this class, how would explain a mutual fund that performs spectacularly in a single year but fails to perform that well in subsequent years?

According to Fama, the strong performance in one year is mostly due to noise ϵ, not true skill. The model g is not consistently close to the true process t, so performance goes back to average over time.

(v) [easy] Did the Manic Momentum model validate? Explain.

No, the Manic Momentum model did not validate. It seemed to work on past data, but when tested in new data it failed, so it was overfit.

(w) [easy] Are stock market bubbles noticable while we're in them? Explain.

Bubbles are hard to notice while we're in them because it's difficult to separate signal from noise, and people tend to follow the crowd. It's much easier to identify them after the fact.

(x) [easy] What is the implication of Shiller's model for a long-term investor in stocks?

Shiller's model suggests that stock returns are somewhat predictable in the long run, so long-term investing is more reliable than trying to predict short-term movements.

(y) [easy] In lecture one, we spoke about "heuristics" which are simple models with high error but extremely easy to learn and live by. What is the heuristic Silver quotes on p358 and why does it work so well?

The heuristic is to stay invested and not try to time the market (I believe day trading is an example of timing the market). It works well because it avoids reacting to short-term noise and reduces mistakes.

(z) [easy] Even if your model at predicting bubbles turned out to be good, what would prevent you from executing on it?

> Even with a good model, it's hard to act on it because there is still uncertainty and risk. You could be early or wrong, and there may be pressure not to go against the market.

(aa) [easy] How can heuristics get us into trouble?

> Heuristics can get us into trouble because they oversimplify reality and ignore important information, which can lead to large errors in certain situations.

Problem 2

These are some questions related to probability estimation modeling. Let X denote the design matrix with n rows and $p+1$ columns (with the first column being $\mathbf{1}_n$ and the other columns being linearly independent predictors) and $\boldsymbol{y}$ is the binary response vector of size n and use this notation throughout your responses.

(a) [easy] What is g_0 if you are modeling probability estimates?

> The null model for probability estimation is the average of the responses: $g_0 = \bar{y}$. So g_0 is just the proportion of 1's in the data.

(b) [easy] What is $\mathcal{H}_{pr}$ for the probability estimation algorithm that employs the linear model in the covariates with logistic link function?

> For logistic regression, the hypothesis set is:
> $\mathcal{H}_{pr} = \left\{\frac{1}{1+e^{-xw}} : w \in \mathbb{R}^{p+1}\right\}$.
> This means we take the linear model xw and pass it through the logistic function.

(c) [easy] What is $\mathcal{H}_{pr}$ for the probability estimation algorithm that employs the linear model in the covariates with cloglog link function?

> For cloglog regression, the hypothesis set is:
> $\mathcal{H}_{pr} = \left\{1 - e^{-e^{xw}} : w \in \mathbb{R}^{p+1}\right\}$.
> This uses a different link function but still starts with xw.

(d) [easy] Let $\mathcal{A} : \boldsymbol{b} = \arg\max_{\boldsymbol{w} \in \mathbb{R}^{p+1}} \{\ldots\}$. Derive the expression that replaces the $\ldots$ which will be a function of $X, \boldsymbol{y}, \boldsymbol{w}, n$. Note: this algorithm fits a "logistic regression".

Logistic regression finds parameters by maximizing the log-likelihood:
$\boldsymbol{b} = \arg\max_{\boldsymbol{w}\in\mathbb{R}^{p+1}} \sum_{i=1}^{n} [y_i \log(\hat{p}_i) + (1-y_i)\log(1-\hat{p}_i)]$ where the hypothesis set $\hat{p}_i = \frac{1}{1+e^{-x_i w}}$
So we get:
$\boldsymbol{b} = \arg\max_{\boldsymbol{w}\in\mathbb{R}^{p+1}} \sum_{i=1}^{n} [y_i(x_i w) - \log(1+e^{x_i w})]$

(e) [easy] Why is logistic regression an example of a "generalized linear model" (glm)?

Logistic regression is a generalized linear model because it starts with a linear model, $x_i.\boldsymbol{b}$, but then applies a link function to turn it into a probability.

(f) [easy] Consider $\boldsymbol{x}_*$ to be a new unit. For its prediction, the probability estimate that $y_* = 1$ is 37%, what is the log odds of $y_* = 1$?

The probability estimate is 37%, so $\hat{p} = 0.37$.
$\log\left(\frac{\hat{p}}{1-\hat{p}}\right) = \log\left(\frac{0.37}{1-0.37}\right) = \log\left(\frac{0.37}{0.63}\right) \approx -0.5322$

(g) [easy] If, $\boldsymbol{x}_*\boldsymbol{b} = 3.1415$ where $\boldsymbol{b}$ is the result of the logistic regression fit, what is the probability estimate that $y_* = 1$?

$\hat{p} = \frac{1}{1+e^{-x^*\boldsymbol{b}}}$
$x^*\boldsymbol{b} = 3.1415$
$\hat{p} = \frac{1}{1+e^{-3.1415}} \approx 0.9586$

(h) [harder] If, $\boldsymbol{x}_*\boldsymbol{b} = 3.1415$ where $\boldsymbol{b}$ is the result of the probit regression fit, what is the probability estimate that $y_* = 1$?

$\hat{p} = \Phi(x^*\boldsymbol{b})$
$x^*\boldsymbol{b} = 3.1415$
$\hat{p} = \Phi(3.1415)$
$\Phi(3.1415) \approx 0.9992$

(i) [easy] In probability estimation modeling, what is the formula for the Brier Score performance metric? Prove the Brier score is always non-positive.

The Brier Score is: $s_i := -(y_i - \hat{p}_i)^2$
The average Brier Score is: $\bar{s} := \frac{1}{n}\sum_{i=1}^{n} -(y_i - \hat{p}_i)^2$
The Brier Score is always non-positive because a square is always non-negative: $(y_i - \hat{p}_i)^2 \geq 0$
So when we put a negative sign in front: $-(y_i - \hat{p}_i)^2 \leq 0$
So, $\bar{s} \leq 0$

(j) [easy] In probability estimation modeling, what is the formula for the Log Scoring Rule performance metric? Prove the Log Scoring Rule is always non-positive.

The Log Scoring Rule is defined as: $s_i := y_i \ln(\hat{p}_i) + (1 - y_i)\ln(1 - \hat{p}_i)$
The average log score is: $\bar{s} := \frac{1}{n}\sum_{i=1}^{n} s_i = \frac{1}{n}\sum_{i=1}^{n}\left(y_i \ln(\hat{p}_i) + (1 - y_i)\ln(1 - \hat{p}_i)\right)$
$0 < \hat{p}_i < 1$, which implies: $\ln(\hat{p}_i) \leq 0 \quad$ and $\quad \ln(1 - \hat{p}_i) \leq 0$
Since $y_i \in \{0, 1\}$, each term $s_i \leq 0$.
Therefore: $\bar{s} \leq 0$

(k) [difficult] Generalize linear probability estimation to the case where $\mathcal{Y} = \{C_1, C_2, C_3\}$, i.e. a nominal variable with $L = 3$ levels.

Assume the logistic link function. Write down the objective function that is argmax'd over the parameters (you define what these parameters are — that is part of the question). Once you get the answer you can see how this easily goes to $L > 3$, an arbirtrary[1] number of response levels.

For $Y = \{C_1, C_2, C_3\}$, we use multinomial logistic regression. Let C_3 be the reference class and define parameters $\boldsymbol{w}_1, \boldsymbol{w}_2 \in \mathbb{R}^{p+1}$.
The probability estimates are: $\hat{p}_{i1} = \frac{e^{x_i \cdot \boldsymbol{w}_1}}{1 + e^{x_i \cdot \boldsymbol{w}_1} + e^{x_i \cdot \boldsymbol{w}_2}}$
$\hat{p}_{i2} = \frac{e^{x_i \cdot \boldsymbol{w}_2}}{1 + e^{x_i \cdot \boldsymbol{w}_1} + e^{x_i \cdot \boldsymbol{w}_2}}$
$\hat{p}_{i3} = \frac{1}{1 + e^{x_i \cdot \boldsymbol{w}_1} + e^{x_i \cdot \boldsymbol{w}_2}}$
Let $I(y_i = C_k)$ be 1 if $y_i = C_k$ and 0 otherwise.
The objective function is:
$\arg\max_{\boldsymbol{w}_1, \boldsymbol{w}_2 \in \mathbb{R}^{p+1}} \sum_{i=1}^{n} \left[I(y_i = C_1)\ln(\hat{p}_{i1}) + I(y_i = C_2)\ln(\hat{p}_{i2}) + I(y_i = C_3)\ln(\hat{p}_{i3})\right]$
So we choose $\boldsymbol{w}_1$ and $\boldsymbol{w}_2$ to make this as large as possible.
ChatGPT help

[1]Note: The algorithm for general L is known as all of the following: "multinomial logistic regression", "polytomous LR", "multiclass LR", "softmax regression", "multinomial logit" (mlogit), the "maximum entropy" (MaxEnt) classifier, and the "conditional maximum entropy model". You can inflate your resume with lots of redundant jazzy terms by doing this one question!

For the next two questions, let $n_1 := \sum \mathbb{1}_{y_i=1}$ and $n_0 := \sum \mathbb{1}_{y_i=0}$ so that $n = n_0 + n_1$. Then assume $n_1 \neq n_0$. This is equivalent to letting $n_1 = cn$ and $n_0 = (1-c)n$ and assuming $c \in [0,1] / \left\{\frac{1}{2}\right\}$.

(l) [harder] [MA] Prove the Brier score is always higher for g_0 vs the model where you set $\hat{p}_i = \frac{1}{2}$ for all i. Hint: $(1-c)c < \frac{1}{2}$ for $c \in [0,1] / \left\{\frac{1}{2}\right\}$.

For the null model g_0, we have $\hat{p}_i = c$, so:
$\bar{s}_{g_0} = \frac{1}{n}\sum_{i=1}^{n} -(y_i - c)^2 = -c(1-c)$
For the model $\hat{p}_i = \frac{1}{2}$:
$\bar{s}_{1/2} = -\frac{1}{4}$
Since $c(1-c) < \frac{1}{4}$ when $c \neq \frac{1}{2}$, we have: $-c(1-c) > -\frac{1}{4}$
So: $\bar{s}_{g_0} > \bar{s}_{1/2}$

(m) [difficult] [MA] Prove the Log Scoring Rule is always higher for g_0 vs the model where you set $\hat{p}_i = \frac{1}{2}$ for all i.

For the null model g_0, we have $\hat{p}_i = c$, so:
$\bar{s}_{g_0} = \frac{1}{n}\sum_{i=1}^{n} [y_i \ln(c) + (1-y_i)\ln(1-c)] = c\ln(c) + (1-c)\ln(1-c)$
For the model $\hat{p}_i = \frac{1}{2}$:
$\bar{s}_{1/2} = \ln(\frac{1}{2})$
$c^c(1-c)^{1-c} > \frac{1}{2}$ for $c \in [0,1] / \left\{\frac{1}{2}\right\}$.
Taking ln of both sides gives:
$\ln\left(c^c(1-c)^{1-c}\right) > \ln(\frac{1}{2})$
So: $c\ln(c) + (1-c)\ln(1-c) > \ln(\frac{1}{2})$
So: $\bar{s}_{g_0} > \bar{s}_{1/2}$
ChatGPT help

Problem 3

These are some questions related to polynomial-derived features and logarithm-derived features in use in OLS regression.

(a) [harder] What was the overarching problem we were trying to solve when we started to introduce polynomial terms into $\mathcal{H}$? What was the mathematical theory that justified this solution? Did this turn out to be a good solution? Why / why not?

We introduced polynomial terms into H to reduce misspecification error. A simple linear hypothesis space, such as $\mathcal{H}_0 = \{w_0 + w_1 x\}$, can underfit if $f(x)$ is not close to linear. So we expanded $\mathcal{H}$ by adding derived features like $x^2, x^3, \ldots$, which gives the model more flexibility. This was done to reduce misspecification error by using a larger candidate set $\mathcal{H}$.
The mathematical theory that justified this was the Weierstrass Approximation Theorem, which says a continuous function on a closed interval can be approximated by a polynomial.
It helped reduce underfitting, but it was not always a good solution because high-degree polynomials can become too flexible, lose interpretability, and overfit the data.

(b) [harder] We fit the following model: $\hat{\boldsymbol{y}} = b_0 + b_1 x + b_2 x^2$. What is the interpretation of b_1? What is the interpretation of b_2? Although we didn't yet discuss the "true" interpretation of OLS coefficients, do your best with this.

b_1 helps determine the slope, and b_2 determines how the slope changes as x changes.

(c) [difficult] Assuming the model from the previous question, if $x \in \mathcal{X} = [10.0, 10.1]$, do you expect to "trust" the estimates b_1 and b_2? Why or why not?

I would not fully trust the separate estimates b_1 and b_2. The range of x is extremely small, so x and x^2 will be very similar over that interval. The model may still predict well in this range, but the coefficients are not trustworthy.

(d) [difficult] We fit the following model: $\hat{\boldsymbol{y}} = b_0 + b_1 x_1 + b_2 \ln(x_2)$. We spoke about in class that b_1 represents loosely the predicted change in response for a proportional movement in x_2. So e.g. if x_2 increases by 10%, the response is predicted to increase by $0.1 b_2$. Prove this approximation from first principles.

$\hat{y} = b_0 + b_1 x_1 + b_2 \ln(x_2)$
Consider a change in x_2 to x_2'. Then:
$\Delta\hat{y} = b_2 \ln(x_2') - b_2 \ln(x_2)$
$= b_2 [\ln(x_2') - \ln(x_2)]$
$= b_2 \ln\left(\frac{x_2'}{x_2}\right)$
If x_2 changes proportionally, then $\frac{x_2'}{x_2} = 1 + r$, so: $\Delta\hat{y} = b_2 \ln(1 + r)$
Using $\ln(1 + r) \approx r$ for small r: $\Delta\hat{y} \approx b_2 r$

(e) [easy] When does the approximation from the previous question work? When do you expect the approximation from the previous question not to work?

The approximation works when r is small (small percentage changes).
It does not work well when r is large, because $\ln(1+r) \approx r$ is no longer accurate.

(f) [harder] We fit the following model: $\ln(\hat{\boldsymbol{y}}) = b_0 + b_1 x_1 + b_2 \ln(x_2)$. What is the interpretation of b_1? What is the *approximate* interpretation of b_2? Although we didn't yet discuss the "true" interpretation of OLS coefficients, do your best with this.

b_1 means a one unit increase in x_1 leads to an approximate proportional change in $\hat{y}$ of b_1.
b_2 means a proportional increase in x_2 leads to an approximate proportional change in $\hat{y}$ of $b_2 r$.

(g) [easy] Show that the model from the previous question is equal to $\hat{\boldsymbol{y}} = m_0 m_1^{x_1} x_2^{b_2}$ and interpret m_1.

$\ln(\hat{y}) = b_0 + b_1 x_1 + b_2 \ln(x_2)$
Exponentiate:
$\hat{y} = e^{b_0 + b_1 x_1 + b_2 \ln(x_2)}$
$= e^{b_0} e^{b_1 x_1} e^{b_2 \ln(x_2)}$
$= e^{b_0} (e^{b_1})^{x_1} x_2^{b_2}$
Let $m_0 = e^{b_0}$ and $m_1 = e^{b_1}$:
$\hat{y} = m_0 m_1^{x_1} x_2^{b_2}$

Problem 4

These are some questions related to extrapolation.

(a) [easy] Define extrapolation and describe why it is a net-negative during prediction.

Extrapolation is when we make predictions for x^* values that are outside the range of the training data X.
It is a net-negative because the model has not seen data in that region, so predictions rely only on assumptions of the model, which may be wrong. This can lead to large errors.

(b) [easy] Do models extrapolate differently? Explain.

Yes, models extrapolate differently because they make different assumptions about the relationship between x and y.
A linear model extrapolates in a straight line, while a polynomial model curves.
So different models can give different predictions outside the data.

(c) [easy] Why do polynomial regression models suffer terribly from extrapolation?

Polynomial regression models suffer from extrapolation because higher-degree polynomials can grow very quickly outside the data range.
Small changes in x can lead to very large changes in $\hat{y}$, making predictions unstable and unreliable.

Problem 5

These are some questions related to the model selection procedure discussed in lecture.

(a) [easy] Define the fundamental problem of "model selection".

The fundamental problem of model selection is choosing a model that performs well on new, unseen data.
We want to pick $g \in \mathcal{H}$ that minimizes prediction error, but we only observe performance on the training data, which can lead to overfitting.

(b) [easy] Using two splits of the data, how would you select a model?

Split the data into training and validation sets.
Fit models on the training set, then evaluate their performance on the validation set.
Select the model that performs best on the validation set.

(c) [easy] Discuss the main limitation with using two splits to select a model.

We use less data for training, which can reduce model accuracy.

(d) [easy] Using three splits of the data, how would you perform model selection?

Split the data into training, validation, and test sets.
Use the training set to fit models, the validation set to select the best model, and the test set to evaluate the final model.

(e) [easy] How does using both inner and outer folds in a double cross-validation nested resampling procedure improve the model selection procedure?

> Inner folds are used to select the best model, while outer folds are used to evaluate performance.
> This improves model selection by giving a more honest estimate of how the model performs on new data.

(f) [easy] Describe how g_{final} is constructed when using nested resampling on three splits of the data.

> After selecting the best model using validation, we refit the model on all available training data.
> This fitted model is g_{final}.

(g) [easy] Describe how you would use this model selection procedure to find hyperparameter values in algorithms that require hyperparameters.

> Treat each hyperparameter setting as a different model in $\mathcal{H}$.
> Use validation (or cross-validation) to compare them and choose the one with the best performance.

(h) [difficult] Given raw features $x_1, \ldots, x_{p_{raw}}$, produce the most expansive set of transformed p features you can think of so that $p \gg n$.

> Create many transformed features such as:
> $x_1^2, x_1^3, \ldots, x_p^2, x_p^3$
> $\ln(x_1), \ln(x_2), \ldots$
> $x_1 x_2, x_1 x_3, \ldots$
> indicator variables and other nonlinear transformations
> This creates a large feature set where $p \gg n$.
> ChatGPT confirmation/help

(i) [easy] Describe the methodology from class that can create a linear model on a subset of the transformed featuers (from the previous problem) that will not overfit.

> Construct a large set of transformed features, then define $\mathcal{H}$ to include only models that use subsets of these features. By restricting $\mathcal{H}$ to simpler models by using fewer features, we reduce model complexity and prevent overfitting even when $p \gg n$.

Problem 6

These are some questions related to the CART algorithms.

(a) [easy] Write down the step-by-step $\mathcal{A}$ for regression trees.

1. Start with all data D.
2. Try all splits of the form $x_j \leq s$.
3. For each split, compute the weighted SSE.
4. Choose the split with the lowest weighted SSE.
5. Split the data into two nodes.
6. Repeat on each node.
7. Stop when nodes are too small.
8. In each leaf, predict the average y.

(b) [difficult] Describe $\mathcal{H}$ for regression trees. This is very difficult but doable. If you can't get it in mathematical form, describe it as best as you can in English.

$\mathcal{H}$ for regression trees is the set of all piecewise constant functions made by recursive axis-aligned splits.
So, $\mathcal{H}$ contains all models that split the feature space X into rectangular regions, then assign one constant prediction to each region.

(c) [harder] Think of another "leaf assignment" rule besides the average of the responses in the node that makes sense.

Use the median instead of the mean.

(d) [harder] Assume the y values are unique in $\mathbb{D}$. Imagine if $N_0 = 1$ so that each leaf gets one observation and its $\hat{\boldsymbol{y}} = y_i$ (where i denotes the number of the observation that lands in the leaf) and thus it's very overfit and needs to be "regularized". Write up an algorithm that finds the optimal tree by pruning one node at a time iteratively. "Prune" means to identify an inner node whose daughter nodes are both leaves and deleting both daughter nodes and converting the inner node into a leaf whose $\hat{\boldsymbol{y}}$ becomes the average of the responses in the observations that were in the deleted daughter nodes. This is an example of a "backwards stepwise procedure" i.e. the iterations transition from more complex to less complex models.

1. Start with the fully grown tree where $N_0 = 1$, meaning each leaf has one observation.
2. Let this be the most complex tree.
3. Find all inner nodes whose two children are both leaves.
4. For each such node, prune it:
- delete the two children
- make the node a leaf
- set $\hat{y}$ to the average of the y values in those two leaves
5. For each possible prune, compute validation error.
6. Choose the prune that gives the lowest validation error.
7. Repeat this process one prune at a time.
8. Continue until only the root node remains.
9. Select the tree with the lowest validation error.

(e) [difficult] Provide an example of an $f(\boldsymbol{x})$ relationship with medium noise δ where vanilla OLS would beat regression trees in oos predictive accuracy. Hint: this is a trick question.

$f(x) = \beta_0 + \beta_1 x$ with medium noise δ.
OLS would beat regression trees because it matches the true linear form exactly. Regression trees approximate with step functions, which can have higher error out-of-sample.

(f) [easy] Write down the step-by-step $\mathcal{A}$ for classification trees. This should be short because you can reference the steps you wrote for the regression trees in (a).

1. Start with all data D.
2. Try all splits of the form $x_j \leq s$.
3. For each split, compute the weighted Gini.
4. Choose the split with the lowest weighted Gini.
5. Repeat on each node.
6. Stop when nodes are too small.
7. In each leaf, predict the majority class.

(g) [difficult] Think of another objective function that makes sense besides the Gini that can be used to compare the "quality" of splits within inner nodes of a classification tree.

Use misclassification error instead of Gini.
For a node, define:
$ME = 1 - \mathbf{max}_\ell \hat{p}_\ell$ #not sure why format is messed up
where $\hat{p}_\ell$ is the proportion of observations in class C_ℓ.
Then for a split, compute the weighted misclassification error and choose the split with the lowest value.